N+ Network Security

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Assignment-4

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1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

5 Common Network Threats

Network Threat	Simple Meaning	Real-World Example
1. Phishing	Fake messages or links are used to steal personal information.	WhatsApp: A scammer sends a fake link to steal a user's login details.
2. DDoS Attack	Attackers send too many requests to a server, making it unavailable.	Instagram: Attackers flood the service with requests so users cannot access it.
3. Man-in-the-Middle	An attacker secretly intercepts communication between two parties.	Paytm: An attacker on unsecured public Wi-Fi tries to intercept user data.
4. Malware	Harmful software that can damage a device or steal information.	Android: A user installs a fake payment app that steals personal information.
5. Password Attack	An attacker tries to guess or steal a user's password.	Instagram: An attacker uses a stolen password to try to access an account.

Conclusion: These threats can steal personal information, damage devices, or stop services from working. We can stay safe by using strong passwords, enabling two-factor authentication, and avoiding suspicious links.

2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.Constraint: The script should print the number of attempts it took to guess the correct PIN.

Python Program: Brute-Force Attack Simulation

```
secret_pin = "1234"
attempts = 0

for i in range(10000):
    guess = str(i).zfill(4)
    attempts += 1

    if guess == secret_pin:
        print("Correct PIN:", secret_pin)
        print("PIN Found:", guess)
        print("Total Attempts:", attempts)
        break
```

Output:

```
Correct PIN: 1234
PIN Found: 1234
Total Attempts: 1235
```

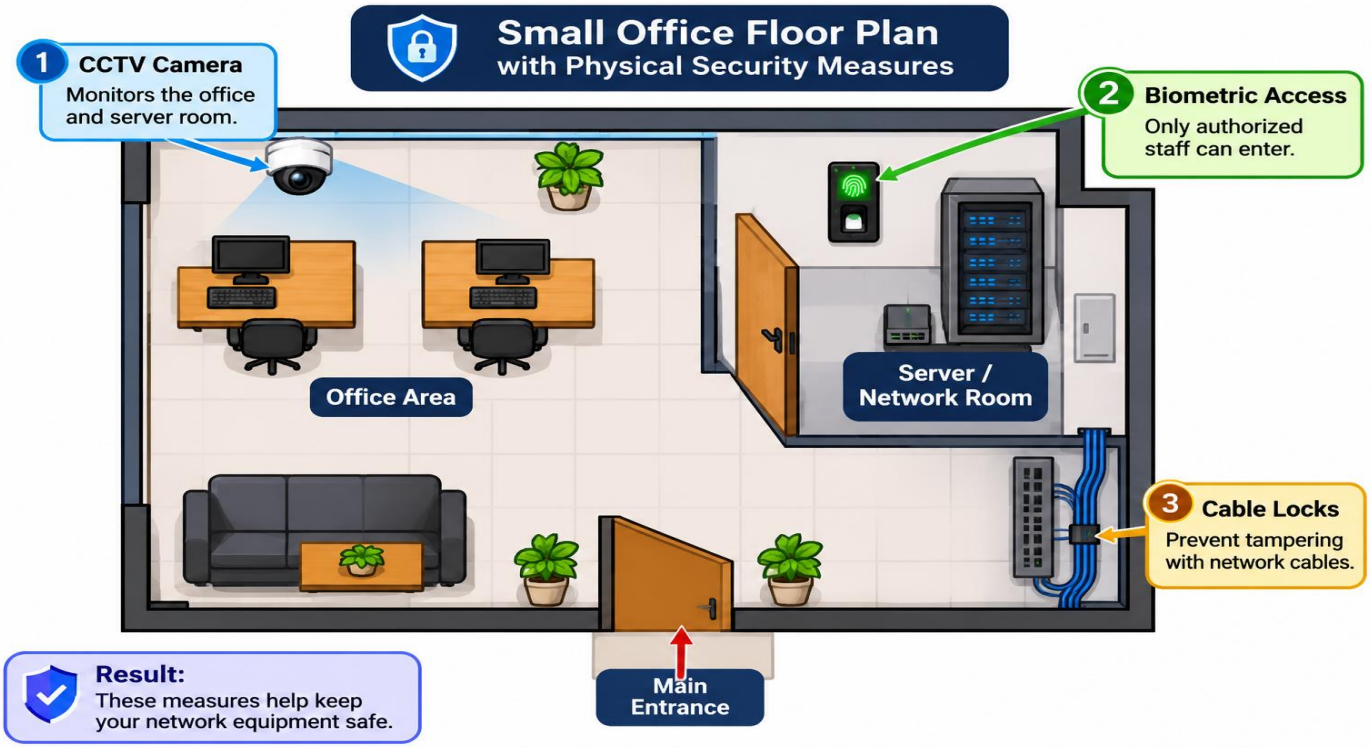
Conclusion: This program tries all 4-digit PIN combinations from 0000 to 9999. It stops when the correct PIN is found and displays the total number of attempts.

3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

Physical Security Risks and Their Solutions

No.	Physical Security Risk	Mitigation (Solution)
1	Fire	Install fire alarms and fire extinguishers.
2	Theft	Install CCTV cameras and door locks.
3	Power Failure	Use UPS and backup generators.
4	Water Leakage	Install water leak detection sensors.

4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).



1. CCTV Camera – Monitors the office and server room.
2. Biometric Access – Only authorized staff can enter the server room.
3. Cable Locks – Prevent unauthorized access to network cables.

5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

Security Checklist for a College Computer Lab

1. Install Antivirus: Install antivirus software on all lab computers to protect against viruses and malware.

2. Use Firewall: Enable firewalls to block unauthorized network traffic.

3. Strong Passwords: Use strong passwords to protect computers and user accounts.

4. Install CCTV Cameras: Install CCTV cameras to monitor the computer lab.

5. Restricted Access: Allow only authorized students and staff to enter the lab.

6. Lock Network Devices: Keep routers, switches, and servers in a locked room.

7. Regular Updates: Update operating systems and software regularly to fix security problems.

Conclusion: These 7 security steps help protect a college computer lab from network attacks and physical security risks.